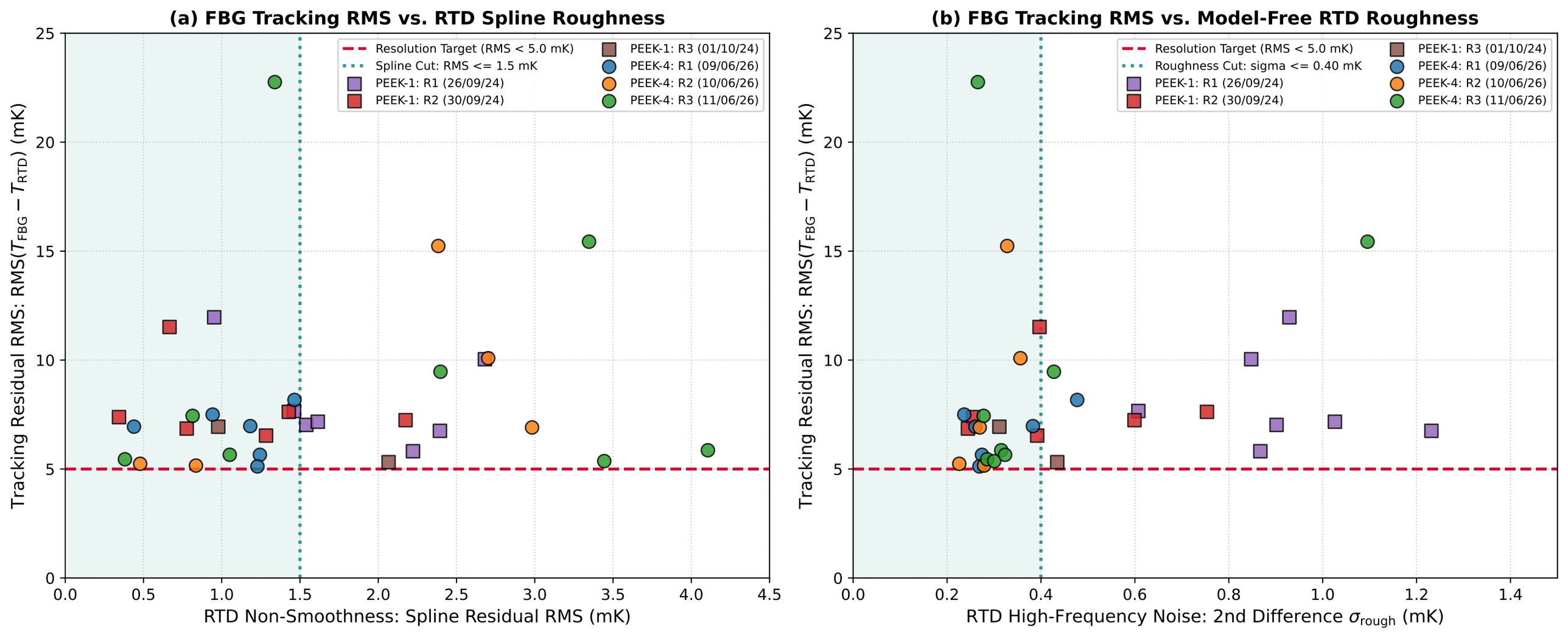
(a) FBG Tracking RMS vs. RTD Spline Roughness
Resolution Target (RMS < 5.0 mK)
Spline Cut: RMS <= 1.5 mK
PEEK-1: R1 (26/09/24)
PEEK-1: R2 (30/09/24)
PEEK-1: R3 (01/10/24)
PEEK-4: R1 (09/06/26)
PEEK-4: R2 (10/06/26)
PEEK-4: R3 (11/06/26)
Tracking Residual RMS: RMS($T_{FBG} - T_{RTD}$) (mK)
RTD Non-Smoothness: Spline Residual RMS (mK)
0
5
10
15
20
25
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
(b) FBG Tracking RMS vs. Model-Free RTD Roughness
Resolution Target (RMS < 5.0 mK)
Roughness Cut: sigma <= 0.40 mK
PEEK-1: R1 (26/09/24)
PEEK-1: R2 (30/09/24)
PEEK-1: R3 (01/10/24)
PEEK-4: R1 (09/06/26)
PEEK-4: R2 (10/06/26)
PEEK-4: R3 (11/06/26)
Tracking Residual RMS: RMS($T_{FBG} - T_{RTD}$) (mK)
RTD High-Frequency Noise: 2nd Difference σ_{rough} (mK)
0
5
10
15
20
25
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4